

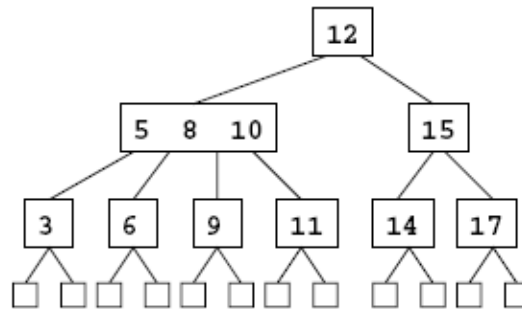
INT202 Complexity of Algorithms

Tutorial 2

Q1. Show the height of a (2, 4) tree storing n keys is $\Theta(\log n)$.

Q2. Let T be a (2, 4) tree shown below, which stores items with integer keys. Draw all the steps and the resulting trees obtained by performing the following operations on the original (given) T .

- Insert an item with key 7 into T .
- Remove an item with the key 17 from T .



Q3. Given the following pre-order and in-order binary tree traversals, draw the binary tree.

Pre-order A, B, C, D, E, F

In-order B, A, D, E, C, F